HARSHINI KODALI

AI/ML Engineer | LLM Systems & RAG | MLOps | AWS / GCP

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SUMMARY

AI/ML Engineer with 3+ years of experience building and deploying machine learning and LLM-powered systems in enterprise environments. Experienced in developing credit risk models, implementing Retrieval-Augmented Generation (RAG) pipelines, and deploying containerized ML services on AWS. Strong background in transformer models, model explainability, and production monitoring across structured and unstructured data applications.

TECHNICAL SKILLS

Languages: Python, SQL, Bash
Machine Learning: Classification, Regression, Forecasting, Feature Engineering, Model Evaluation, Scikit-learn, XGBoost, LightGBM
Deep Learning: PyTorch, TensorFlow, Transformer Models, Hugging Face Transformers, Fine-Tuning (LoRA/PEFT)
LLM & Retrieval Systems: Retrieval-Augmented Generation (RAG), LangChain, LlamaIndex, Prompt Engineering, Embeddings, Semantic Search, Vector Databases (FAISS), RAG Evaluation (RAGAS), FinBERT
API & Deployment: FastAPI, REST APIs, Docker, Kubernetes
Cloud & Distributed Systems: AWS (S3, EC2, SageMaker, Lambda), GCP (Vertex AI), Apache Spark
Data Processing: Pandas, NumPy, ETL Pipelines, Feature Stores
Databases: PostgreSQL, MongoDB (NoSQL)
MLOps & Experimentation: MLflow, Airflow, Model Monitoring, SHAP, CI/CD, A/B Testing, Git

EXPERIENCE

AI/ML Engineer - *Simmons Bank* - Houston, TX *Jan 2025 – Dec 2025*

- Designed and deployed a Retrieval-Augmented Generation (RAG) system using LangChain and FAISS over 100K+ financial documents, reducing manual compliance review time by 40%.
- Developed credit risk prediction models using XGBoost and PyTorch, improving model AUC from 0.78 to 0.87 through feature engineering and hyperparameter tuning.
- Built LLM-powered semantic search and multi-document summarization workflows using embeddings to enable contextual retrieval across structured and unstructured financial data.
- Containerized and deployed ML services using FastAPI and Docker on AWS (EC2, S3, SageMaker) with Kubernetes orchestration for scalable inference.
- Automated model retraining pipelines using Airflow and MLflow, reducing model refresh cycles by 30% and improving deployment reliability.
- Implemented SHAP-based explainability and model monitoring to support regulatory compliance and model governance.

AI/ML Engineer - *Cognizant Technology Solutions* - Chennai, India *Jan 2023 – Dec 2023*

- Architected end-to-end machine learning pipelines for enterprise clients, delivering classification and forecasting models that improved operational efficiency by 22% across customer analytics workflows.
- Spearheaded feature engineering and model selection initiatives using Scikit-learn and XGBoost, increasing prediction precision by 18% through systematic cross-validation and performance tuning.
- Implemented NLP-driven automation solutions leveraging BERT-based transformer models to extract insights from unstructured text datasets, reducing manual processing effort by 35%.
- Collaborated with software engineering teams to integrate ML services into REST-based microservices architecture, ensuring seamless API consumption and production readiness.
- Automated data ingestion and preprocessing workflows using Spark and Airflow, improving pipeline scalability and reducing batch processing time by 28%.
- Evaluated and benchmarked early LLM use cases for internal knowledge retrieval and document summarization, contributing to proof-of-concept deployments for enterprise AI initiatives.

Associate ML Engineer - *Mindtree Ltd.* - Bangalore, India *Sep 2021 – Dec 2022*

- Analyzed large structured datasets to develop supervised learning models for customer segmentation and churn prediction, increasing campaign targeting accuracy by 20%.
- Engineered data preprocessing pipelines to address missing values, class imbalance, and feature scaling, improving model recall by 15% across validation datasets.
- Constructed ensemble models including Random Forest and XGBoost to enhance predictive stability and reduce overfitting in production scoring systems.
- Conducted exploratory data analysis and statistical validation to uncover behavioral patterns, supporting business decisions.
- Assisted in deploying ML models as RESTful services in collaboration with backend teams, enabling real-time inference capabilities for internal applications.
- Documented model assumptions, evaluation metrics, and performance benchmarks to ensure reproducibility and alignment with project stakeholders.

KEY PROJECTS

- **RAG Evaluation Framework:** Built an evaluation framework using RAGAS metrics to benchmark RAG pipelines across 1,000+ question-answer pairs, enabling systematic prompt and retrieval optimization.
- **Transformer Fine-Tuning for Financial NER:** Fine-tuned FinBERT for financial named-entity recognition using parameter-efficient techniques, achieving F1 score of 0.91 on regulatory clause extraction tasks.

EDUCATION

M.S. in Data Science - University of Houston, TX *Dec 2025*
B. Tech in Computer Science & Engineering - Vel Tech University, India *May 2023*